

Homework Assignment 12
Econ 629 - Texas A&M University
Fall 2012

Posted: Thursday, Nov. 29th, 2012.
Due: Tuesday, Dec. 4th (Class time).

1. Problem 9 in Section 4.4 in Lecture notes.
2. Suppose that there are two inputs with prices (w_1, w_2) and one output with price p . Consider a Cobb-Douglas production function $f : \mathbb{R}_+^2 \rightarrow \mathbb{R}$, $f(z_1, z_2) = Az_1^\alpha z_2^\beta$.
 - (a) $Y^f = ?$
 - (b) Characterize the combinations of α and β for which Y^f satisfies NI, ND, and CR of scale.
 - (c) Calculate the associated supply correspondence, profit function, conditional factor demand correspondence and cost function.
 - (d) Verify Hotelling's Lemma for each (w_1, w_2, p) where it holds.